

HUMAIRA ANZUM NEHA

Lecturer of CSE at UIU || M.Sc. Student at BRAC University || Graduate of KUET

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Faculty Profile Google Scholar LinkedIn GitHub Portfolio

EDUCATION

2024–Present	M.Sc. in Computer Science and Engineering CGPA: 4.00/4.00; 6 credits completed	BRAC University
2019–2024	B.Sc. in Computer Science and Engineering CGPA: 3.87/4.00	Khulna University of Engineering & Technology (KUET)
2018	Higher Secondary Certificate GPA: 5.00/5.00	Viqarunnisa Noon School and College
2016	Secondary School Certificate GPA: 5.00/5.00	Viqarunnisa Noon School and College

RESEARCH INTERESTS

Human–Centered AI, Mutlimodal AI, Human–Computer Interaction, Computer Vision, Natural Language Processing, Large Language Models, Accessibility

SKILLS

Programming:	Python, C/C++, Java, JavaScript, SQL
AI & ML:	PyTorch, OpenCV, Hugging Face Transformers, Scikit-Learn
Web & Mobile:	HTML, CSS, Tailwind CSS, ASP.NET, Firebase, Android Studio
Database:	MySQL, Oracle
Development Tools:	Git, GitHub, LaTeX, OpenGL

WORK EXPERIENCE

2024–Present	Lecturer in Department of CSE United International University - I am currently serving as a Full-time Lecturer in the Dept. of CSE at United International University. - Courses Taught: Database Management Systems, Data Mining, Object Oriented Programming, Computer Architecture, Data Structures and Algorithms.
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RESEARCH EXPERIENCE

2024 1st author	Automated Object Identification for the Visually Impaired  - Developed an edge-compatible assistive system for common-object detection, Bangladeshi currency and denomination recognition, and face recognition. - Collected and annotated custom image datasets for model training and evaluation: object-detection dataset and Bangladeshi currency dataset. - Supervisor: Dr. Md. Sheikh Sadi, Professor of CSE, KUET. - Technologies Used: Python, PyTorch, OpenCV, YOLOv5.  Repository
2024	An Assistive Device for the Visually Impaired for Enhanced Navigation and Reading  - Developed embedded spectacle glasses with real-time obstacle avoidance, fall detection, and textual information extraction modules. - Supervisor: Dr. Md. Sheikh Sadi, Professor of CSE, KUET. - Technologies Used: Python, PyTorch, OpenCV, YOLOv5, Raspberry Pi, OCR, Accelerometer, and ToF Distance Sensor.

2023 1st author	Mobile Application to Collect Data and Measure Blood Component Levels in a Non-invasive Way - Developed a mobile application for non-invasive measurement of blood components. - The user places their fingertip on an NIR LED device, and a 15-second video is recorded. The video is analyzed to extract PPG signals, which are processed through a DNN model to predict glucose and hemoglobin levels. - Supervisor: S M Taslim Uddin Raju, Lecturer of CSE, KUET (on leave). - Technologies Used: Kotlin, Android Studio, Python, Photoplethysmography (PPG) Signal Analysis, and OpenCV.
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OPEN DATASETS AND RESEARCH RESOURCES

2026	MusicCaps-Curated-Tags Curated MusicCaps annotations into 216 standardized labels across genre, instrument, mood, vocal, and tempo categories using an LLM-assisted pipeline with manual review and validation.
2024	Computer Vision Datasets Assistive Object Detection Dataset (approximately 21,000 YOLO-annotated images across eight classes) and Bangladeshi Currency Classification Dataset.

ACADEMIC PROJECTS

Python, Pygame	Abalone AI Developed a playable Abalone board game in Pygame with an AI opponent powered by Minimax and alpha-beta pruning.
Bootstrap, JavaScript, ASP.NET	Smart&Safe Developed a web app for buying and selling used smartphones with identity verification (NID + security information) and automated IMEI screening through an admin dashboard.
Java, Firebase, Android Studio	Together Created a task and deadline management mobile application with collaboration and communication features for teams and coworkers.

ACHIEVEMENTS

3 times	Dean's List Award Awarded for consistent academic excellence and strong semester-wise academic performance.	Khulna University of Engineering & Technology (KUET)
2021	Top-10 Finalist Designed and prototyped a community-based application idea selected among the hackathon's top 10 entries.	UNDP Women's Digital Innovation Hackathon 2021
2018–2019 to 2021–2022	Top-3 CGPA (among 120 students) Ranked in the top 3 by cumulative academic performance within the cohort.	Khulna University of Engineering & Technology (KUET)

REFERENCES

1	Dr. Sk. Mohammad Masudul Ahsan Professor Department of Computer Science and Engineering Khulna University of Engineering & Technology Bangladesh	✉ smahsan@cse.kuet.ac.bd
2	Dr. Muhammad Sheikh Sadi Professor Department of Computer Science and Engineering Khulna University of Engineering & Technology Bangladesh	✉ sadi@cse.kuet.ac.bd